

GLU#

<https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.GLU.html>

Description:

Applies the gated linear unit function. $\text{GLU}(a, b) = a \otimes \sigma(b)$ where a is the first half of the input matrices and b is the second half. dim (int) – the dimension on which to split the input. Default: -1 Input: $(\ast 1, N, \ast 2)$ where $\ast$ means, any number of additional dimensions Output: $(\ast 1, M, \ast 2)$ where $M = N/2$

Function Signature

```
class torch.nn.modules.activation.GLU(dim=-1)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> m = nn.GLU()
>>> input = torch.randn(4, 2)
>>> output = m(input)
```

Detailed Documentation

Documentation

Applies the gated linear unit function.

$GLU(a,b)=a\otimes\sigma(b)$ where a is the first half of the input matrices and b is the second half.

`dim (int)` – the dimension on which to split the input. Default: -1

Input: $(\ast_1,N,\ast_2)$ where $\ast$ means, any number of additional dimensions

Output: $(\ast_1,M,\ast_2)$ where $M=N/2$

Return the extra representation of the module.

Runs the forward pass.